

06-node-express

06. Node.js & Express + Database

Level: □ Intermediate

Jam: 9 (4 minggu x 2 sesi + 1 sesi database)

Prasyarat: JavaScript Fundamentals, TypeScript Basics

Output: REST API production-ready + PostgreSQL

Tujuan

- Paham runtime Node.js
- Bikin REST API pake Express + TypeScript
- Middleware, routing, error handling
- Koneksi database PostgreSQL
- CRUD operations

Contoh Express API

```
import express from 'express';
const app = express();
app.use(express.json());

interface Note { id: number; title: string; content: string; }
const notes: Note[] = [];

app.get('/api/notes', (req, res) => res.json(notes));

app.post('/api/notes', (req, res) => {
  const { title, content } = req.body;
  const note: Note = { id: notes.length + 1, title, content };
  notes.push(note);
  res.status(201).json(note);
});

app.listen(3000, () => console.log('API di :3000'));
```

Database SQL (PostgreSQL)

Basic SQL

```
CREATE TABLE users (  
  id SERIAL PRIMARY KEY,  
  name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) UNIQUE NOT NULL,  
  created_at TIMESTAMP DEFAULT NOW()  
);  
  
INSERT INTO users (name, email) VALUES ('Budi', 'budi@email.com');  
SELECT * FROM users;  
SELECT name, email FROM users WHERE id = 1;  
UPDATE users SET name = 'Andi' WHERE id = 1;  
DELETE FROM users WHERE id = 1;
```

Koneksi dari Node.js

```
import { Pool } from 'pg';  
const pool = new Pool({ connectionString: process.env.DATABASE_URL });  
  
app.get('/api/users', async (req, res) => {  
  const result = await pool.query('SELECT * FROM users');  
  res.json(result.rows);  
});
```